

Adaptive Queue Length + Fuzzy Logic Congestion Control in Clustered WSN

This document describes the design for congestion control in a clustered Wireless Sensor Network (WSN). The solution combines **adaptive queue length adjustment** at Cluster Heads (CHs) and **independent fuzzy logic-based congestion detection** using Active Queue Management.

1. Overview of the Proposed Design

Our system contains two independent components:

A. Adaptive Queue Length Mechanism (AQLM)

- Queue length at each CH is *not fixed*.
- It is dynamically adjusted based on:
 - **Packet Loss Ratio (PLR)**
 - **Downstream/Neighbour Cluster Head Capacity**
- AQLM runs **only when PLR exceeds a threshold**.

B. Fuzzy Logic-Based Congestion Detection (FIS + AQM)

- Runs periodically, independent of AQLM.
- Inputs include:
 - Average Queue Length (AQL)
 - Packet Loss Ratio (PLR)
 - Arrival Rate
 - Residual Energy
 - Packet Priority
 - Current Queue Capacity
- Outputs:
 - Drop Probability
 - Throttling/Rate Adjustment
 - Congestion Notification Flag

The two systems **do not control each other internally**, but AQLM affects the queue state which the fuzzy engine reads as an input.

2. Workflow of the System

Step 1: Measure metrics periodically

- PLR updated via EWMA
- AQL calculated from queue occupancy
- Neighbouring CH capacity obtained via control adverts

Step 2: If $PLR > \text{threshold}$, adapt queue length

- Compute desired queue length using PLR
- Scale desired queue using downstream capacity
- Apply smoothing/hysteresis to avoid oscillation
- Update the queue size

Step 3: Fuzzy Logic runs independently

- Reads current queue size and other inputs
 - Produces AQM actions
 - Handles congestion detection + response
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3. Control/Advert Packet Format (Lightweight)

Neighbour CHs periodically advertise their capabilities using a small, piggybacked message.

Advert Packet Structure

```
{  
  CH_ID:      uint8,  
  service_rate: float (EWMA of pkts/sec),  
  avail_buffer: uint8 (free queue slots),  
  energy_level: uint8 (optional),  
  timestamp:  uint32  
}
```

- Sent every $T_{adv} = 2\text{--}5$ seconds
- Missing advert for $T_{timeout}$ $\Rightarrow$ assume low capacity

Use in algorithm

- `service_rate` used to compute downstream capacity
 - `avail_buffer` used as optional buffer-awareness factor
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4. Adaptive Queue Length Algorithm (AQLM)

Parameters

- Q_{min}, Q_{max} – allowed queue limits
 - PLR_{th} – threshold at which adaptation begins
 - C_{ref} – reference send rate of the CH
 - β – smoothing factor
 - ΔQ_{min} – minimum effective change
 - $step_{max}$ – max queue adjustment per update
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5. Fuzzy Inference System (FIS) — Runs Independently

Inputs

- Average Queue Length (AQL)
- Packet Loss Ratio (PLR)
- Arrival Rate
- Packet Priority
- Residual Energy
- **Current Queue Capacity** (output of AQLM)

Outputs

- Drop Probability
- Transmission Rate Adjustment
- Congestion Notification (ECN flag)

Operation

- Runs every **T_FIS** seconds
 - Reads the *current* queue size, but does not influence its calculation
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7. Advantages of the Design

- **Independent, modular controls** (robust and easy to tune)
 - **Stable queue adaptation** avoids oscillation
 - **Neighbour-aware** adjustment prevents congestion cascades
 - FIS focuses purely on **detecting and responding** to congestion
 - Adaptation triggers only when needed (high PLR)
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8. Conclusion

This design provides a novel, hybrid congestion control mechanism for clustered WSNs by:

- Dynamically adapting queue length based on PLR and neighbour capacity
- Detecting congestion using a separate fuzzy logic AQM engine
- Ensuring stability, energy efficiency, and non-cascading operation